gemstone-py Type-Safe Smalltalk Codegen

Generating typed wrappers from Protocol classes

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Assets are local SVG illustrations stored in the repository.

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Type-Safe Smalltalk Codegen

`gemstone-codegen` turns registered Python `Protocol` classes into concrete `TypedOop` wrappers. The generated files are ordinary Python modules, so IDEs can autocomplete methods and your application code no longer has to spell the same Smalltalk strings at every call site.

Install

The generator ships with `gemstone-py`:

```
python -m pip install gemstone-py
gemstone-codegen --help
```

For source checkouts:

```
python -m pip install -e ".[examples,dev]"
python -m gemstone_py.codegen --help
```

Define Protocols

Create a module with one or more `@gemstone_class(...)` protocols:

```
from typing import Protocol
from gemstone_py import gemstone_class, gemstone_selector

@gemstone_class("OkzBooking", async_=True)
class OkzBookingProto(Protocol):
    status: str

    @classmethod
    @gemstone_selector("findById:")
    def find_by_id(cls, booking_id: str) -> "OkzBookingProto":
        ...

    def mark_paid(self, at_posix_seconds: int) -> None:
        ...

    def yourself(self) -> "OkzBookingProto":
        ...

    def customer(self) -> "OkzCustomerProto":
        ...
```

```
    @gemstone_selector("transferTo:byUserId:")
    def transfer(self, user_id: int, by_user_id: int) -> None:
        ...

@gemstone_class("OkzCustomer", async_=True)
class OkzCustomerProto(Protocol):
    name: str
```

`async_=True` asks the generator to emit both sync and async wrappers.

Generate Wrappers

Run the generator from the repository or application root:

```
gemstone-codegen \
  --module examples.typed_access.codegen_demo.models \
  --output examples/typed_access/codegen_demo/generated \
  --clean
```

The output is meant to be checked in. That keeps reviews, editor indexing, and type checking predictable:

```
git add examples/typed_access/codegen_demo/generated
```

The generated package includes `.pyi` stubs and `py.typed` so type checkers treat the checked-in wrapper package as typed. `--clean` removes stale generated wrapper modules and stale wrapper stubs when a Protocol is renamed or deleted.

The generator keeps runtime code lightweight. It emits ordinary Python wrapper classes that call the existing session API; it does not add a registry, identity map, object cache, or global mapper.

Each generated wrapper class also exposes lightweight runtime metadata:

```
print(OkzBooking.__gemstone_protocol__)
print(OkzBooking.__gemstone_selectors__["find_by_id"])
```

Use that metadata for diagnostics, generated-wrapper audits, and UI tooling without parsing generated source files.

Use package generation for protocols that return other generated protocols. Single-class `generate_wrapper(...)` only has enough context to resolve self-typed returns.

Use `--check` in CI or pre-commit hooks to catch drift:

```
gemstone-codegen \
  --module examples.typed_access.codegen_demo.models \
  --output examples/typed_access/codegen_demo/generated \
  --check
```

The repository CI runs the same check through:

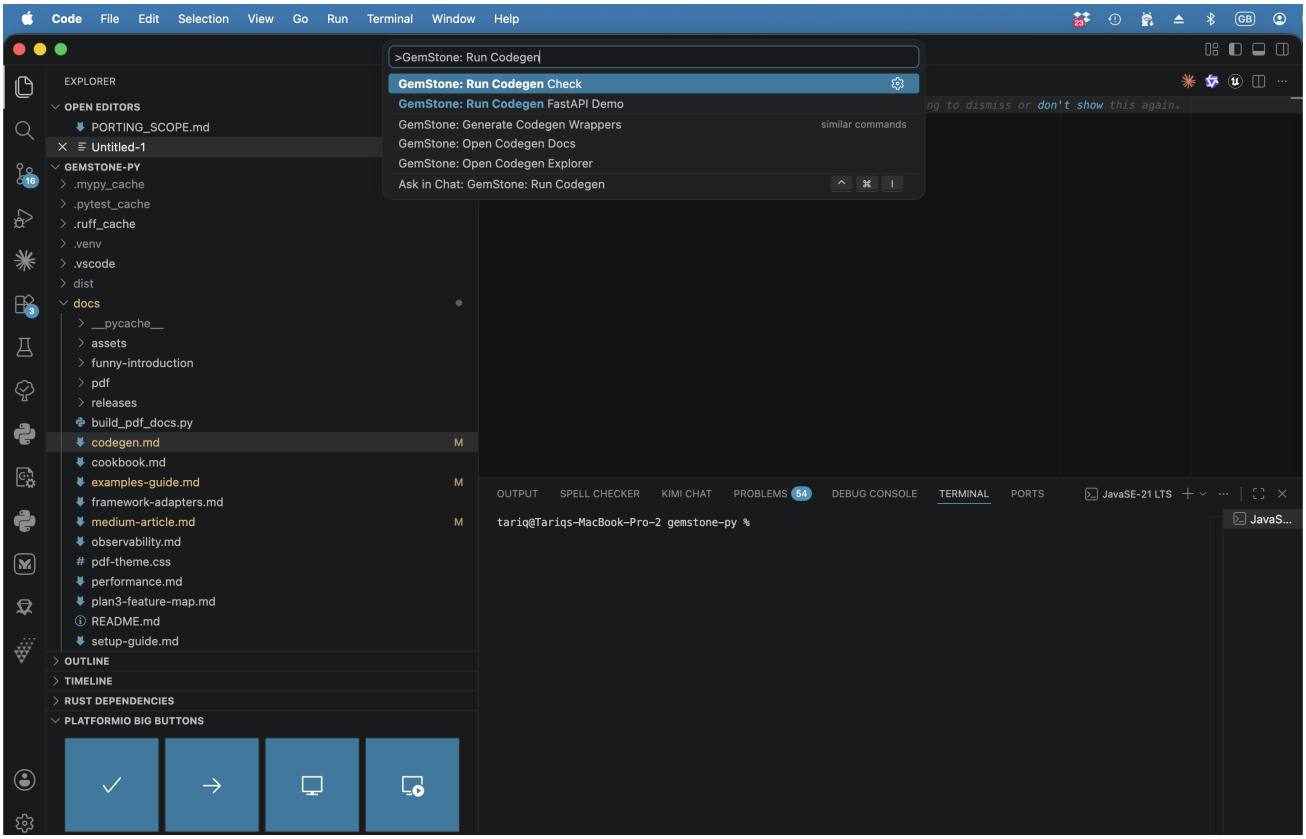
```
./scripts/check_codegen.sh
```

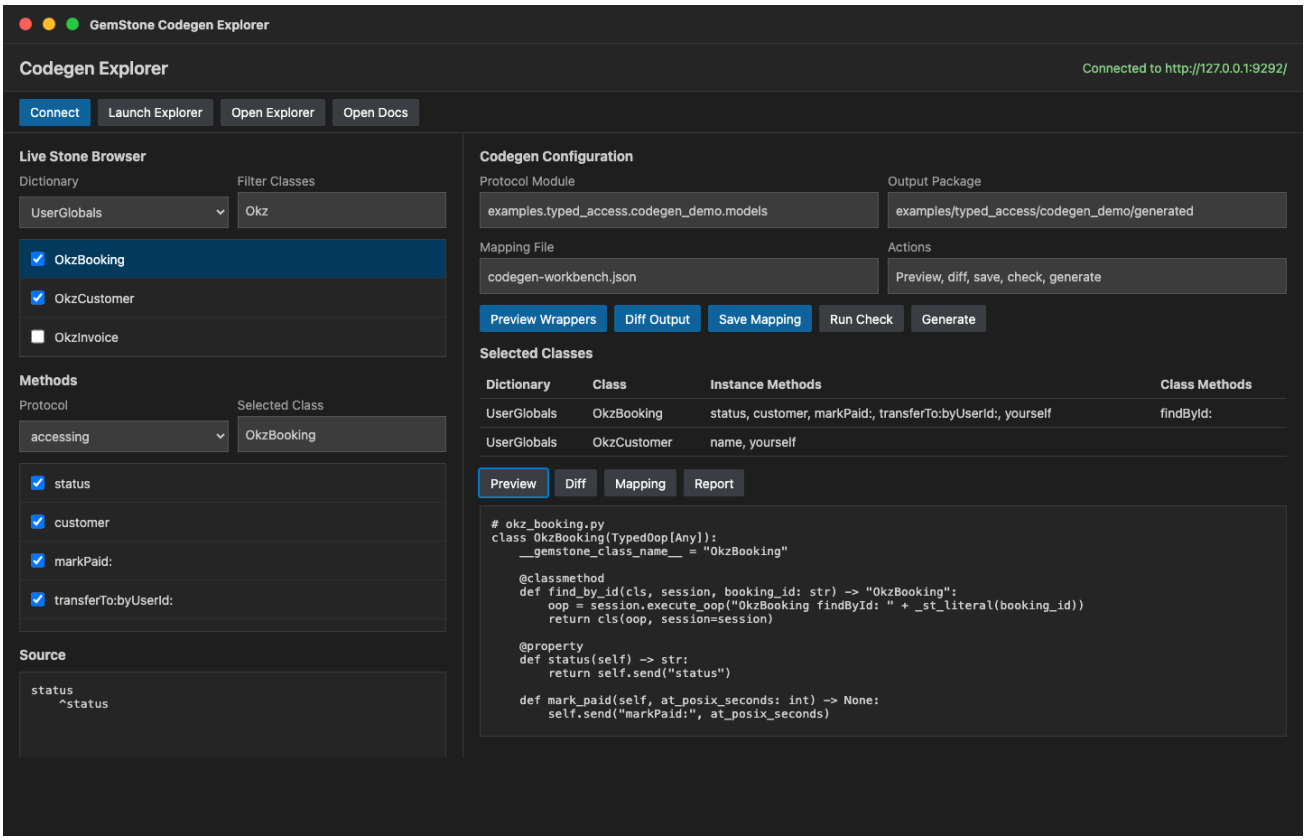
If your application uses `pre-commit`, it can consume the packaged hook:

```
repos:
  - repo: https://github.com/unicompute/gemstone-py
    rev: main
    hooks:
      - id: gemstone-codegen-check
```

Visual Codegen Explorer

The VS Code workbench includes `GemStone: Open Codegen Explorer` for the visual workflow around the same generator. It sits between the live `python-gemstone-database-explorer` class browser and the checked-in Protocol module. The browser explorer provides the stable `/codegen/*` JSON API; the VS Code workbench consumes that API instead of owning GemStone discovery or code-generation rules.





1. Start the database explorer with `gemstone-py: Launch Database Explorer`.
2. Run `GemStone: Open Codegen Explorer`.
3. Click `Connect` to load dictionaries from the live stone.

4. Browse dictionaries, classes, protocols/categories, methods, and source

   through `/codegen/dictionaries`, `/codegen/classes`, `/codegen/protocols`, `/codegen/methods`, and `/codegen/source`.

5. Select classes and instance/class-side methods as wrapper targets.
6. Save or load normalized selection JSON with fields, selectors, generated

   Python names, argument names, and return annotations.

7. Use `Preview Wrappers` to call `/codegen/preview` before writing files.
8. Use `Diff Output` to compare the preview output against the configured

   generated package.

9. Run `Run Check`, `Generate`, or `Run Demo` when the preview looks right.

The visual selection file is intentionally separate from the Protocol module. It records the live classes, fields, methods, class-side selectors, and edited Python metadata you chose while you design or review wrappers; the generated Python still comes from explicit `@gemstone_class(...)` Protocol definitions so the API remains reviewable and type-checkable. The database explorer normalizes this file with `/codegen/export-selection`, so the browser UI, VS Code workbench, and future tooling can share the same handoff format.

The demo mapping file is:

```
examples/typed_access/codegen_demo/codegen-workbench.example.json
```

It records this useful first selection in the current normalized schema:

Class	Fields	Class-side selectors	Instance selectors
`OkzBooking`	`status`	`findById:`	`customer`, `markPaid:`, `transferTo:byUserId:`, `yourself`
`OkzCustomer`	`name`	none	`yourself`

Use that mapping as the checklist when you browse the live classes, or import it into the explorer to restore the richer method metadata. Keep the Protocol module as the source of truth for the generated Python API.

Concrete Demo Workflow

The repository includes a complete, reviewable Codegen demo under `examples/typed_access/codegen_demo/`.

Preview generation without touching checked-in files:

```
python -m examples.typed_access.codegen_demo.preview
python -m examples.typed_access.codegen_demo.preview --show-source
```

Regenerate the checked-in package:

```
gemstone-codegen \
  --module examples.typed_access.codegen_demo.models \
  --output examples/typed_access/codegen_demo/generated \
  --clean
```

Verify that generated files are current:

```
gemstone-codegen \
  --module examples.typed_access.codegen_demo.models \
  --output examples/typed_access/codegen_demo/generated \
  --check
```

Run the generated-wrapper FastAPI demo:

```
python -m examples.typed_access.codegen_demo.run --reload
```

Probe the generated sync wrapper against a live stone:

```
export GS_STONE_NAME=gs64stone
export GS_USERNAME=DataCurator
export GS_PASSWORD=swordfish
python -m examples.typed_access.codegen_demo.live_probe --booking-id B-1001
```

Selector Mapping

Default mapping is intentionally small:

Python shape	Generated Smalltalk selector
`status`	`status`
`find_by_id(id)`	`findById:`
`mark_paid(at)`	`markPaid:`

Python shape	Generated Smalltalk selector
`transfer_to(user_id, by_user_id)`	`transferTo:byUserId:`

For selectors that do not map cleanly, use `@gemstone_selector(...)`. The explicit selector must have the same keyword count as the Python method has arguments.

Return Mapping

The generator uses simple Protocol annotations to choose the send path and the generated Python return type:

Protocol return annotation	Generated behaviour
no annotation	`perform_value(...)` / `session.eval(...)`, returns `Any`
`None`	sends the message and returns `None`
same Protocol class, wrapper class, or `Self`	uses `perform_oop(...)` or `execute_oop(...)` and wraps the returned OOP
another generated Protocol in the same module	uses `perform_oop(...)` or `execute_oop(...)` and lazily imports the target wrapper
simple builtins such as `str`, `int`, `float`, `bool`	value send with that return annotation

That means this method returns another typed wrapper:

```
def yourself(self) -> "OkzBookingProto":
    ...
```

and this method returns a generated `OkzCustomer` or `AsyncOkzCustomer` wrapper:

```
def customer(self) -> "OkzCustomerProto":
    ...
```

while this method is treated as a mutating command:

```
def mark_paid(self, at_posix_seconds: int) -> None:
    ...
```

Argument Conversion

Generated class-side wrappers turn simple Python literals into Smalltalk source only for the small supported set: strings, integers, floats, booleans, and `None`. Instance-side generated wrappers send raw OOP arguments through `perform_*`, so pass an existing raw OOP, `Oop` marker, or wrapper-managed OOP when the selector expects an object reference.

Keep complex query expressions, dynamic Smalltalk, and application-specific value conversion outside generated source. For scalar-ish Python values such as `datetime`, `date`, `Decimal`, or `UUID`, use the explicit converter registry described in the User Manual before passing the resulting `Oop` marker to the wrapper or session call.

Sync Usage

Generated class-side methods take a session and build the Smalltalk source:

```
from gemstone_py import GemStoneConfig, GemStoneSession
from examples.typed_access.codegen_demo.generated import OkzBooking

with GemStoneSession(config=GemStoneConfig.from_env()) as session:
    booking = OkzBooking.find_by_id(session, "B-1001")
    print(booking.status)
    booking.mark_paid(1_779_912_000)
    same_booking = booking.yourself()
    customer = booking.customer()
    print(customer.name)
```

That class-side call evaluates:

```
OkzBooking findById: 'B-1001'
```

Instance methods call `perform_value(...)` for value returns and `perform_oop(...)` for self-typed wrapper returns through the session associated with the returned `TypedOop`.

Async Usage

When the Protocol is registered with `async_=True`, the generator also emits an `Async...` wrapper:

```
from fastapi import Depends, FastAPI
from gemstone_py import GemStoneConfig
from gemstone_py.aio import AsyncSession
from gemstone_py.aio.fastapi import session_dependency
from examples.typed_access.codegen_demo.generated import AsyncOkzBooking
```

```
app = FastAPI()
get_gemstone =
session_dependency(config=GemStoneConfig.from_env(require_credentials=False))

@app.get("/bookings/{booking_id}")
async def booking_status(
    booking_id: str,
    session: AsyncSession = Depends(get_gemstone),
) -> dict[str, str]:
    booking = await AsyncOkzBooking.find_by_id(session, booking_id)
    return {"status": str(await booking.status())}
```

The repository includes a runnable version of that handler:

```
python -m examples.typed_access.codegen_demo.run --reload
```

With the server running, open:

```
http://127.0.0.1:8000/
http://127.0.0.1:8000/docs
http://127.0.0.1:8000/bookings/B-1001
```

The root and docs routes are local FastAPI checks. The booking route requires GemStone credentials and a reachable stone because it evaluates `OkzBooking findById:`.

Current Scope

The first generator pass handles:

- annotated fields and `@property` methods as no-argument sends
- instance methods as `perform_value(...)` or `perform_oop(...)` sends based on

  return annotations

- class methods as class-side Smalltalk source strings
- self-returning class and instance methods as wrapped `TypedOop` results
- same-module cross-Protocol returns as lazily imported generated wrappers
- string, integer, float, boolean, and `None` literals in class-side calls
- conservative argument conversion that rejects unsupported source literals

  instead of guessing

- explicit selector overrides with `@gemstone_selector(...)`
- checked-in sync and async generated modules with `.pyi` stubs, `py.typed`,

  stale-file cleanup, and CI/pre-commit drift checks

- runtime source-Protocol and selector-map metadata on every generated wrapper
- a runnable FastAPI demo route backed by the generated async wrapper
- an opt-in live smoke test for generated wrappers against GemStone `Date`

It does not marshal arbitrary Python objects into class-side source literals or replace hand-written Smalltalk for complex queries. Use it for method-shaped object access and keep raw `session.eval(...)` or `SmalltalkBridge` calls where a full Smalltalk expression is clearer.

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